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ORIGINAL ARTICLE

# Validation of Proposed Criteria for Progressive Pulmonary Fibrosis

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## Abstract

**Rationale:** Criteria for progressive pulmonary fibrosis (PPF) have been proposed, but their prognostic value beyond categorical decline in FVC remains unclear.

**Objectives:** To determine whether proposed PPF criteria predict transplant-free survival (TFS) in patients with non–idiopathic pulmonary fibrosis (IPF) forms of interstitial lung disease (ILD).

**Methods:** A retrospective, multicenter cohort analysis was performed. Patients with diagnoses of fibrotic connective tissue disease–associated ILD, fibrotic hypersensitivity pneumonitis, and non-IPF idiopathic interstitial pneumonia from three U.S. centers and one UK center constituted the test and validation cohorts, respectively. Cox proportional hazards regression was used to test the association between 5-year TFS and ⩾10% FVC decline, followed by 13 additional PPF criteria satisfied in the absence of ⩾10% FVC decline.

**Measurements and Main Results:** One thousand three hundred forty-one patients met the inclusion criteria. A ⩾10% relative FVC decline was the strongest predictor of reduced TFS and showed consistent TFS association across cohorts, ILD subtypes, and treatment groups, resulting in a phenotype that closely resembled IPF. Ten additional PPF criteria satisfied in the absence of 10% relative FVC decline were also associated with reduced TFS in the U.S. test cohort, with 6 maintaining TFS associations in the UK validation cohort. Validated PPF criteria requiring a combination of physiologic, radiologic, and symptomatic worsening performed similarly to their stand-alone components but captured a smaller number of patients.

**Conclusions:** An FVC decline of ⩾10% and six additional PPF criteria satisfied in the absence of such decline identify patients with non-IPF ILD at increased risk for death or lung transplantation.

**Keywords:** idiopathic interstitial pneumonia; interstitial lung disease; idiopathic pulmonary fibrosis; progressive fibrosing interstitial lung disease; progressive pulmonary fibrosis

(Received in original form January 18, 2022; accepted in final form August 9, 2022)

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‡These authors contributed equally to this work.

Supported by NHLBI grants T32HL007013 (J.V.P.), K23HL146942 (A.A.), T32HL007605 (C.T.L.), R01HL093096 (C.K.G.), K23HL148498 (C.A.N.), K23HL138190 (J.M.O.), and R56HL158935 (J.M.O.); National Center for Advancing Translational Sciences grant UL1TR001105 (C.A.N.); and an Action for Pulmonary Fibrosis Mike Bray fellowship (P.L.M.).

Author Contributions: Study design, J.V.P., A.A., C.A.N., P.L.M., and J.M.O.; data collection, J.V.P., A.A., Z.W., C.T.L., A.S., S.G., V.V., E.A.R., A.U.W., C.K.G., F.C., C.A.N., P.L.M., and J.M.O.; data analysis, J.M.O.; interpretation of results, J.V.P., A.A., Z.W., C.T.L., E.A.R., A.U.W., C.K.G., F.C., C.A.N., P.L.M., and J.M.O.; manuscript preparation, J.V.P., A.A., C.A.N., P.L.M., and J.M.O. All authors reviewed and approved of the final manuscript.

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This article has a related editorial.

This article has an online supplement, which is accessible from this issue's table of contents at www.atsjournals.org.

Am J Respir Crit Care Med Vol 207, Iss 1, pp 69–76, Jan 1, 2023
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Originally Published in Press as DOI: 10.1164/rccm.202201-0124OC on August 9, 2022
Internet address: www.atsjournals.org

## At a Glance Commentary

**Scientific Knowledge on the Subject:** A subset of patients with non–idiopathic pulmonary fibrosis (IPF) forms of interstitial lung disease (ILD) display poor survival similar to those with IPF. Criteria for identifying patients with this progressive pulmonary fibrosis phenotype have been proposed, but few have been validated in the target population.

**What This Study Adds to the Field:** This study shows that a relative decline in FVC of ⩾10% is the strongest predictor of reduced transplant-free survival in patients with non-IPF ILD and that the resulting phenotype displayed survival similar to IPF. We also found that six additional progressive pulmonary fibrosis criteria satisfied in the absence of ⩾10% FVC decline were associated with reduced transplant-free survival but that survival after satisfying each criterion was highly heterogeneous depending on the ILD subtype.

Progressive pulmonary fibrosis (PPF) is a devastating condition characterized by high healthcare utilization and poor survival (1–3). Although nearly all patients with idiopathic pulmonary fibrosis (IPF) progress, large but variable proportions of those with non-IPF forms of interstitial lung disease (ILD) also develop PPF, including fibrotic hypersensitivity pneumonitis (fHP), connective tissue disease–associated ILD (CTD-ILD), and non-IPF idiopathic interstitial pneumonia (IIP) (4). Clinical trials (5–7), expert perspectives (8, 9), and a recent international consensus statement (10) have proposed criteria for identifying PPF among patients with non-IPF ILD, including stand-alone measures of lung function decline and combinations of physiologic, radiologic, and symptomatic worsening.

Although potentially helpful for standardizing PPF classification, most proposed PPF criteria were selected on the basis of their known associations with outcomes in patients with IPF (11). Few of these criteria have been validated in the target population of patients with non-IPF ILD, however. One notable exception is a ⩾10% relative decline in FVC, which strongly predicts reduced survival across common ILD subtypes (12–19) and is associated with the highest rate of subsequent FVC decline in PPF clinical trials (20). This reality has implications when applying other proposed PPF criteria, as only those criteria predictive of clinically relevant endpoints in the absence of concurrent FVC decline of ⩾10% are likely to provide value. We recently found that a substantial proportion of patients satisfy more than one PPF criterion concurrently (4), calling into question the number of patients who meet PPF criteria in the absence of ⩾10% FVC decline. Before proposed PPF criteria are adopted, a better understanding of the prognostic value and resultant phenotype identified by these criteria is needed.

In this multicenter investigation, we tested proposed PPF criteria and their stand-alone components for association with 5-year transplant-free survival (TFS) in patients with non-IPF ILD. These criteria included ⩾10% relative FVC decline and 13 additional PPF criteria satisfied in the absence of ⩾10% FVC decline. We hypothesized that few PPF criteria would be associated with reduced TFS in the absence of ⩾10% FVC decline. A test and validation approach was used, with three U.S. centers and one UK center constituting the test and validation cohorts, respectively.

## Methods

### Study Cohorts

This study was performed at the University of California, Davis (UCD), the University of Chicago (UC), the University of Texas Southwestern (UTSW), and Royal Brompton Hospital (RBH), with U.S.-based centers serving as the test cohort and RBH serving as the validation cohort. All patients provided informed consent to participate in site-specific ILD registries (UCD #928979, UC #13-1180/#14163A, UTSW #092017-007/#082010-127, and RBH #19/LO/1879). Consecutive patients ⩾18 years of age with multidisciplinary diagnoses of CTD-ILD (including systemic sclerosis, rheumatoid arthritis, idiopathic inflammatory myopathy, Sjögren's syndrome, systemic lupus erythematosus, mixed connective tissue disease, and vasculitis), fHP, or non-IPF IIP (including idiopathic nonspecific interstitial pneumonia and unclassifiable ILD) and undergoing ILD program evaluation at UCD (January 2014 to December 2017), UC (January 2006 to December 2017), UTSW (January 2006 to December 2017), and RBH (January 2010 to December 2014) were identified. Patients with at least one pulmonary function test (PFT) performed after the baseline PFT were eligible for inclusion. Patients were excluded when *1*) fibrotic extent was <10% on baseline chest computed tomography (CT), as determined by site radiologist, or *2*) baseline FVC and $D_{LCO}$ were unavailable. Patients who received diagnoses of IPF over the same time frame were also identified using site-specific ILD registries to serve as a reference cohort.

### Proposed PPF Criteria

Fourteen proposed criteria, including their stand-alone components, were applied to all patients using available clinical data. These included ⩾10% relative FVC decline and 13 addition criteria (5–10) satisfied in the absence of ⩾10% FVC decline:

1. A 5–9% relative FVC decline (stand-alone component of combined criteria below)
2. A 5–9% absolute FVC decline (6, 7)
3. A ⩾10% absolute $D_{LCO}$ decline (stand-alone component of combined criteria below)
4. A ⩾15% relative $D_{LCO}$ decline (9)
5. CT progression of fibrosis (stand-alone component of combined criteria below)
6. A 5–9% relative FVC decline and worsening respiratory symptoms (5, 8)
7. A 5–9% absolute FVC decline and worsening respiratory symptoms (10)
8. A 5–9% relative FVC decline and ⩾15% relative $D_{LCO}$ decline (8)
9. A ⩾10% absolute $D_{LCO}$ decline and worsening respiratory symptoms (10)
10. CT progression of fibrosis and worsening respiratory symptoms (5, 8, 10)
11. CT progression of fibrosis and 5–9% relative decline in FVC (5, 8, 9)
12. CT progression of fibrosis and 5–9% absolute FVC decline (10)
13. CT progression of fibrosis and ⩾10% absolute decline in $D_{LCO}$ (10)

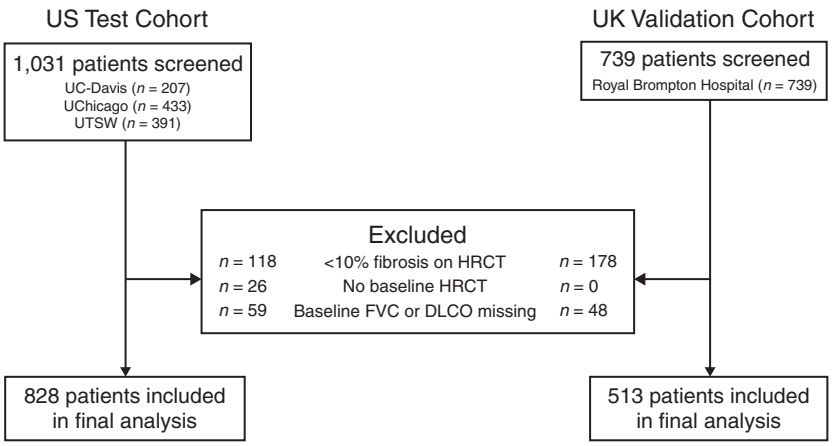


**Figure 1.** Strengthening the Reporting of Observational Studies in Epidemiology diagram. HRCT = high-resolution computed tomography; UC = University of California; UChicago = University of Chicago; UTSW = University of Texas Southwestern.

**Table 1.** Baseline Characteristics for U.S. Test and UK Validation Cohorts

| Characteristic | U.S. Cohort (*n* = 828) | UK Cohort (*n* = 513) |
|---|---|---|
| Center, *n* (%) | | |
| University of California, Davis | 175 (21.1) | N/A |
| University of Chicago | 381 (46.0) | N/A |
| University of Texas Southwestern | 272 (32.9) | N/A |
| Royal Brompton Hospital | 0 (0) | 513 (100) |
| Age, y, mean ± SD | 61.9 ± 12.4 | 58.7 ± 12.6 |
| Male sex, *n* (%) | 369 (44.6) | 202 (39.4) |
| Race and ethnicity, *n* (%) | | |
| White | 572 (69.1) | 380 (74.1) |
| Black | 154 (18.6) | 20 (3.9) |
| Hispanic | 67 (8.1) | 0 (0) |
| Asian | 30 (3.6) | 113 (22.0) |
| Other or unknown | 5 (0.6) | 0 (0) |
| Ever-smoker, *n* (%) | 408 (49.3) | 321 (63.6) |
| ILD classification, *n* (%) | | |
| fHP | 228 (27.5) | 130 (25.3) |
| CTD-ILD | 321 (38.8) | 195 (38.0) |
| RA-ILD | 85 (10.3) | 41 (8.0) |
| SSc-ILD | 81 (9.8) | 63 (12.3) |
| IIM-ILD | 73 (8.8) | 38 (7.4) |
| Other CTD-ILD | 82 (9.9) | 53 (10.3) |
| IIP (non-IPF) | 279 (33.7) | 188 (36.7) |
| Idiopathic NSIP | 98 (11.8) | 92 (17.9) |
| Unclassifiable ILD | 181 (21.9) | 96 (18.7) |
| Pulmonary function, mean ± SD | | |
| FVC % predicted | 66.6 ± 19.4 | 61.5 ± 17.4 |
| $D_{LCO}$ % predicted | 41.4 ± 16.4 | 36.8 ± 12.4 |
| HRCT pattern, *n* (%) | | |
| Definite or probable UIP | 227 (27.4) | 59 (11.5) |
| Indeterminate for UIP | 110 (13.3) | 106 (20.7) |
| Alternative diagnosis | 491 (59.3) | 348 (67.8) |
| Nonspecific interstitial pneumonia | 296 (35.8) | 209 (40.7) |
| Airway-centric fibrosis | 187 (22.6) | 93 (18.1) |
| Treatment exposure, *n* (%) | | |
| Immunosuppressant therapy | 416 (50.2) | 369 (71.9) |
| Antifibrotic therapy | 79 (9.5) | 23 (4.5) |

*Definition of abbreviations*: CTD-ILD = connective tissue disease–associated interstitial lung disease; fHP = fibrotic hypersensitivity pneumonitis; HRCT = high-resolution computed tomography; IIM = idiopathic inflammatory myopathy; IIP = idiopathic interstitial pneumonia; ILD = interstitial lung disease; IPF = idiopathic pulmonary fibrosis; N/A = not applicable; NSIP = nonspecific interstitial pneumonia; OP = organizing pneumonia; RA = rheumatoid arthritis; SSc = systemic sclerosis; UIP = usual interstitial pneumonia.

The date each criterion was first satisfied was determined using longitudinal PFT databases and electronic medical records at each center. Measured values for FVC and $D_{LCO}$ were used to determine relative decline to mitigate any variability in reference equations between cohorts. Categorical $D_{LCO}$ decline was considered present when unable to perform this maneuver. CT progression of fibrosis was considered present when "worsening" or "progression" of fibrosis or ILD was noted on the radiology report. Criteria requiring combinations of physiologic, radiologic, and symptomatic worsening were considered satisfied when each component of a criterion was documented within 365 days of the other (10).

### Statistical Analysis

The primary endpoint assessed was 5-year TFS, defined as the time from baseline PFT to death, lung transplantation, or censoring. PPF criteria were applied over a 4-year period (through July 1, 2019), which captured the majority of follow-up PFTs (*see* Figure E1 in the online supplement) and observed progression events (*see* Figure E2) and provided at least 1 year of follow-up to capture events after the last PPF criterion was satisfied. TFS was compared between groups using multivariable Cox proportional hazards regression adjusted for the gender, age, and physiology ILD index (21). To account for immortal time bias, PPF criteria were modeled as time-dependent covariates. With this approach, observation times before and after satisfying a PPF criterion were considered independent observations (22, 23), with the gender, age, and physiology ILD index at the start of each period used for multivariable adjustment.

After combining cohorts, heterogeneity in the association between TFS and PPF criteria for key subgroups was assessed by fitting the model with a criterion-by-subgroup interaction term and conducting a global chi-square test to determine whether the TFS association for each PPF criterion varied between subgroups. ILD-stratified TFS was then compared with an IPF reference cohort after diagnosis and plotted using the Kaplan-Meier estimator. Survival was censored August 1, 2021. All models were checked to ensure that the proportional hazards assumption was satisfied. Statistical significance was set to $P \leq 0.0036$ for each cohort, with Bonferroni correction for 14 intraclass tests. All statistical analyses

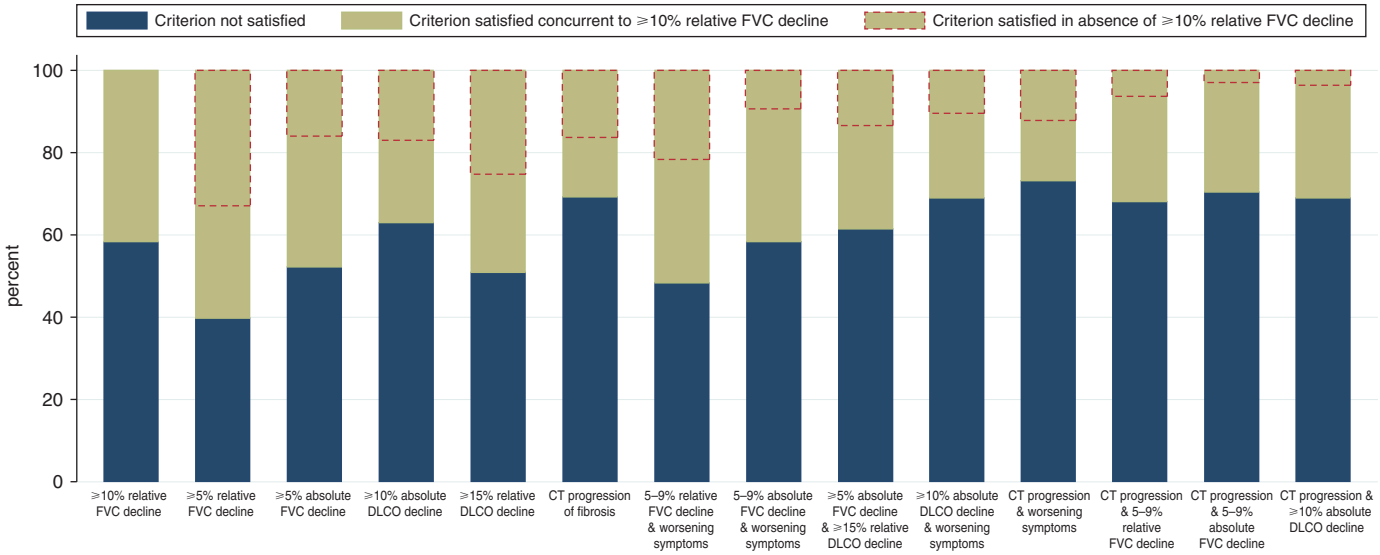


**Figure 2.** Proportion of patients satisfying progressive pulmonary fibrosis features over a 5-year follow-up period with and without concurrent ⩾10% relative FVC decline. CT = computed tomography.

were performed using Stata Release 16 (StataCorp).

## Results

### Study Cohorts

Of 1,031 eligible patients in the U.S. cohort and 739 eligible patients in the UK cohort, 828 and 513 were included in the final analysis (Figure 1). CTD-ILD was the most common ILD subtype, accounting for roughly 40% of each cohort, followed by non-IPF IIP and fHP (Table 1). U.S. cohort patients were older than UK cohort patients (61.9 vs. 58.7 yr), with higher proportions of men and non-White patients. Patients had moderate lung function impairment at the time of enrollment, with those in the U.S. cohort having higher mean percentage predicted FVC and $D_{LCO}$ compared with the UK cohort. Most patients in each cohort had nonspecific interstitial pneumonia on chest imaging, while a definite or probable usual interstitial pneumonia pattern was observed in 27% of U.S. patients and 11.5% of UK patients. A higher proportion of patients in the UK cohort received immunosuppressant therapy, while a higher proportion of patients in the U.S. cohort received antifibrotic therapy. The median time to the first and last follow-up PFT was 5.2 months (interquartile range, 3.3–8.2 mo) and 37.8 months (interquartile range, 19.5–66.2 mo), respectively. TFS was similar between cohorts, with CTD-ILD cohorts demonstrating the best overall survival, followed by fHP and IIP cohorts, which had similar survival (*see* Figure E3). Survival for all three non-IPF ILD cohorts was favorable compared with the IPF reference cohorts (*see* Figure E3).

### Four-Year Incidence of PPF Features

Within 4 years of evaluation, roughly half of patients experienced ⩾10% FVC decline (Figure 2). Among PPF criteria satisfied in the absence of ⩾10% FVC decline, a 5–9% relative FVC decline was most common, followed by a ⩾15% relative $D_{LCO}$ decline (Figure 2). Substantially smaller proportions of patients satisfied PPF criteria in the absence of ⩾10% relative FVC decline, with 50% or more of patients experiencing ⩾10% FVC decline concurrent with most other PPF criteria. Notably, fewer patients satisfied PPF criteria that required combinations of physiologic, radiologic, and symptomatic worsening compared with the stand-alone features constituting these criteria (Figure 2).

### TFS after Satisfying PPF Criteria

An FVC decline of ⩾10% was the strongest predictor of reduced TFS across the U.S. and UK cohorts, with greater than threefold increased risk of death or transplantation in combined analysis (hazard ratio, 3.11; 95% confidence interval, 2.51–3.85) (Table 2). In the absence of ⩾10% FVC decline, 10 of 13 PPF criteria predicted reduced TFS in the U.S. test cohort, with 6 maintaining TFS associations in the UK validation cohort (Table 2). Of validated PPF criteria, three were stand-alone features (5–9% relative FVC decline, ⩾15% relative $D_{LCO}$ decline, and CT progression of fibrosis) and three comprised combinations of physiologic, radiologic, and symptomatic worsening (Table 2). PPF criteria requiring combinations of features showed similar TFS associations compared with stand-alone components constituting these criteria but captured a smaller number of patients (Table 2). Interaction testing revealed no significant heterogeneity in the TFS association for PPF criteria between the U.S. and UK cohorts (Table 2). In combined cohort analysis, all validated PPF criteria showed consistent TFS association across ILD subtypes, with little heterogeneity in TFS association on the basis of interaction analysis (*see* Table E1). Similar results were observed when stratifying the cohort according to immunosuppressant (mycophenolate mofetil, azathioprine, cyclophosphamide, or rituximab) exposure (*see* Table E2) and antifibrotic (nintedanib or pirfenidone) exposure (*see* Table E3).

TFS was significantly better among non-IPF ILD subtypes compared with IPF after diagnosis (Figure 3A) but closely approximated that of IPF after ⩾10% FVC decline (Figure 3B; *see* Table E4). In the absence of ⩾10% FVC decline, TFS was highly variable depending on the ILD subtype and validated PPF criterion satisfied.

**Table 2.** Risk of Death or Lung Transplantation after Satisfying Proposed Progressive Pulmonary Fibrosis Criteria in U.S. Test and UK Validation Cohorts

| PPF Criterion | U.S. Cohort ($n = 828$) | | | UK Cohort ($n = 513$) | | | | Combined Cohort |
|---|---|---|---|---|---|---|---|---|
| | *n* | HR (95% CI) | *P* Value | *n* | HR (95% CI) | *P* Value | *P* Value* | HR (95% CI) |
| ⩾10% relative FVC decline | 404/828 | 3.11 (2.37–4.08) | <0.001 | 241/513 | 3.34 (2.35–4.73) | <0.001 | 0.702 | 3.11 (2.51–3.85) |
| Excluding those with concurrent ⩾10% relative FVC decline | | | | | | | | |
| 5–9% relative FVC decline | 291/577 | 2.87 (2.04–4.03) | <0.001 | 183/366 | 2.36 (1.53–3.63) | <0.001 | 0.428 | 2.58 (1.98–3.35) |
| 5–9% absolute FVC decline | 146/535 | 2.50 (1.70–3.68) | <0.001 | 80/328 | 2.02 (1.22–3.33) | 0.006 | 0.461 | † |
| ⩾10% absolute $D_{LCO}$ decline | 203/624 | 1.93 (1.41–2.65) | <0.001 | 43/394 | 0.91 (0.45–1.81) | 0.779 | 0.057 | † |
| ⩾15% relative $D_{LCO}$ decline | 253/611 | 2.28 (1.65–3.13) | <0.001 | 116/366 | 2.19 (1.43–3.35) | <0.001 | 0.893 | 2.20 (1.71–2.83) |
| CT progression of fibrosis | 135/524 | 1.81 (1.27–2.59) | 0.001 | 62/324 | 2.43 (1.45–4.08) | 0.001 | 0.577 | 1.99 (1.49–2.66) |
| 5–9% relative FVC decline and worsening symptoms | 190/574 | 2.68 (1.94–3.72) | <0.001 | 133/356 | 2.14 (1.41–3.27) | <0.001 | 0.420 | 2.42 (1.87–3.12) |
| 5–9% absolute FVC decline and worsening symptoms | 86/541 | 2.41 (1.58–3.67) | <0.001 | 44/323 | 1.89 (1.01–3.52) | 0.046 | 0.542 | † |
| 5–9% relative FVC decline and ⩾15% relative $D_{LCO}$ decline | 129/596 | 2.40 (1.70–3.40) | <0.001 | 67/368 | 2.29 (1.42–3.70) | <0.001 | 0.725 | 2.29 (1.73–3.02) |
| ⩾10% absolute $D_{LCO}$ decline and worsening symptoms | 126/648 | 1.85 (1.32–2.59) | <0.001 | 29/396 | 0.87 (0.37–2.00) | 0.736 | 0.094 | † |
| CT progression of fibrosis and worsening symptoms | 93/525 | 2.26 (1.56–3.28) | <0.001 | 51/324 | 2.62 (1.54–4.48) | <0.001 | 0.961 | 2.32 (1.72–3.14) |
| CT progression of fibrosis and 5–9% relative FVC decline | 50/493 | 1.63 (0.99–2.70) | 0.055 | 25/302 | 1.33 (0.62–2.85) | 0.467 | 0.402 | † |
| CT progression of fibrosis and 5–9% absolute FVC decline | 27/487 | 1.93 (1.01–3.71) | 0.047 | 10/293 | 1.30 (0.38–4.39) | 0.675 | 0.439 | † |
| CT progression of fibrosis and ⩾10% absolute $D_{LCO}$ decline | 35/486 | 2.13 (1.23–3.67) | 0.007 | 9/288 | 1.00 (0.23–4.29) | 0.988 | 0.324 | † |

*Definition of abbreviations*: CI = confidence interval; CT = computed tomography; HR = hazard ratio; PPF = progressive pulmonary fibrosis.
The analysis involving CT progression excluded those for whom follow-up CT was not performed. The analysis involving symptomatic worsening excluded those for whom change in symptoms could not be ascertained. Estimates adjusted for the gender, age, and physiology interstitial lung disease index at the time of analysis entry.
*Test of whether transplant-free survival association for PPF criteria varied by cohort.
†Not performed because of absence of transplant-free survival association across derivation and validation cohorts at $P < 0.0036$, with Bonferroni adjustment for 14 tests.

After 5–9% relative FVC decline (alone or with worsening respiratory symptoms), TFS approximated IPF in those with non-IPF IIP but remained favorable among those with fHP and CTD-ILD (Figure 4). TFS in those with fHP and non-IPF IIP otherwise closely approximated IPF after satisfying all other validated PPF criteria but remained favorable in those with CTD-ILD (Figures 4B–4E; *see* Table E4).

## Discussion

In this study, we validated previous findings showing that ⩾10% relative FVC decline strongly predicts reduced survival in the non-IPF ILD population (15–18, 24), with consistent findings across cohorts and ILD subtypes. We also found that in the absence of ⩾10% relative FVC decline, six additional PPF criteria were associated with reduced TFS, including three stand-alone features (5–9% relative FVC decline, ⩾15% relative $D_{LCO}$ decline, and CT progression of fibrosis) and three that required combinations of physiologic, radiologic, and symptomatic worsening. Criteria requiring combined features showed similar TFS prediction compared with their stand-alone constituent parts but in a smaller number of patients. When comparing TFS with an IPF reference cohort, we found that a ⩾10% FVC decline resulted in an IPF-like phenotype irrespective of ILD subtype, whereas other validated PPF criteria resulted in an IPF-like phenotype only among those with fHP and non-IPF IIP. To our knowledge, this study is the largest performed in patients with non-IPF ILD to date and the first to validate proposed PPF criteria as they relate to survival.

The PPF construct stemmed from the observation that a substantial proportion of patients with non-IPF ILD develop an IPF-like phenotype, characterized by relentless lung function decline and early mortality (1, 2, 8, 25, 26). Several clinical trials sought to define PPF as part of inclusion criteria, with overlapping but slightly different definitions (5–7). Although some used stand-alone PPF criteria such as categorical FVC decline, others used combinations of worsening respiratory symptoms, lung function, and radiologic fibrosis. In light of these overlapping definitions, a diagnostic guideline defining PPF was recently published by the American Thoracic Society, European Respiratory Society, Japanese

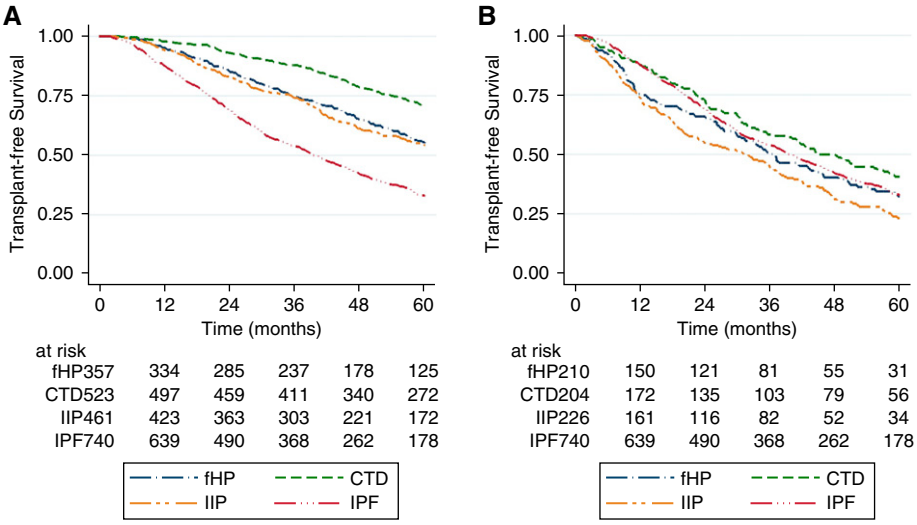


**Figure 3.** TFS among ILD subtypes. Although favorable compared with idiopathic pulmonary fibrosis (IPF) following diagnosis (*A*), TFS among fibrotic hypersensitivity pneumonitis, CTD-ILD, and non-IPF IIP closely approximated IPF after ⩾10% relative FVC decline (*B*). fHP = fibrotic hypersensitivity pneumonitis; CTD = connective tissue disease; IIP = idiopathic interstitial pneumonia; ILD = interstitial lung disease; TFS = transplant-free survival.

Respiratory Society, and Latin American Thoracic Society to standardize PPF classification (10). Although this guideline represents a step in the right direction for standardizing definition of PPF, our results highlight several areas that deserve consideration before broadly implementing these newly released criteria.

First, our data corroborate recent studies suggesting that phenotypic variability remains among ILD subtypes after satisfying proposed PPF criteria (4, 27). Although the risk of death or transplantation after satisfying validated PPF criteria was similar among ILD subtypes compared with those who failed to satisfy these criteria, observed TFS after satisfying validated criteria varied considerably by ILD subtype. Those the CTD-ILD cohort maintained a survival advantage after satisfying all validated PPF criteria except ⩾10% relative FVC decline. Although similar TFS among those with fHP and non-IPF IIP supports aggregating these conditions according to a shared PPF phenotype, phenotypic variability observed with CTD-ILD argues against a "one-size-fits-all" approach to PPF classification when the intent is to create non-IPF ILD cohorts with a homogeneous phenotype. Instead, our data suggest that patients with CTD-ILD may require a different set of criteria to ensure such homogenization. This variability may also extend to individual CTD-ILDs, as outcomes are far from homogeneous in these patients (4, 28). Without homogenization of the PPF phenotype, overrepresentation of one ILD subtype could bias results and reduce generalizability.

Second, our data strongly suggest that a relative decline in FVC of ⩾10% should serve as the linchpin of PPF criteria, as this stand-alone criterion captures a phenotype most like IPF and was the strongest and most consistent predictor of reduced TFS in this large non-IPF ILD cohort. Because ⩾10% relative FVC decline overlapped considerably with other PPF features, with smaller proportions of patients satisfying PPF criteria in the absence of ⩾10% relative FVC, our data suggest that ⩾10% relative FVC decline should be viewed as the primary PPF criterion, with other validated criteria considered secondary.
A recent *post hoc* analysis of the INBUILD (Efficacy and Safety of Nintedanib in Patients With Progressive Fibrosing Interstitial Lung Disease) trial support this approach, as ⩾10% relative FVC decline was the most common feature satisfied in qualifying for trial enrollment and resulted in the most progressive phenotype (5). These findings also have implications when applying criteria included in the recent PPF guideline (10), which would consider a patient with ⩾10% relative FVC decline to have PPF only if concurrent symptomatic or radiographic worsening were present. Although it is likely that most patients with ⩾10% relative FVC decline will experience symptomatic or radiologic worsening, our data suggest that a stand-alone decline in FVC of ⩾10% should be considered a marker of ILD progression irrespective of whether symptomatic or radiologic worsening are present.

Third, our data suggest that although PPF criteria requiring combinations of physiologic, radiologic, and symptomatic worsening are likely to capture a phenotype at increased risk of death or lung transplantation, they will do so at the expense of sensitivity. Stand-alone components of these combined criteria, including 5–9% FVC decline, ⩾15% $D_{LCO}$ decline, and CT progression of fibrosis, effectively identified patients with reduced TFS but did so in a larger number of patients, with only modest differences in TFS association. These differences were driven primarily by missing CT or symptomatic data, which is unlikely to affect clinical trial cohorts but highly likely to affect registry cohorts such as those studied here. This trade-off between sensitivity and specificity for a progressive phenotype will need to be weighed depending on the study in which these criteria are applied.

Next, although we showed that ⩾15% relative $D_{LCO}$ decline was a validated predictor of reduced TFS, there remains concern that isolated $D_{LCO}$ decline may at times represent worsening pulmonary vasculopathy rather than ILD progression. Like ILD progression, pulmonary vasculopathy leads to reduced survival and should be considered when patients meet this criterion in the absence of concurrent FVC decline (29, 30). In addition to ILD progression, an isolated decline in $D_{LCO}$ could also indicate worsening pulmonary hypertension, which has also been linked to reduced survival (31).

Last, the context in which PPF criteria are applied is important. The criteria validated here were tested using 5-year TFS as the primary endpoint. Although measures of ILD survival are commonly assessed when conducting clinical investigation, this outcome is rarely used to gauge therapeutic efficacy in ILD clinical trials. As such, PPF criteria that best predict ILD survival may not optimally identify patients for enrollment in ILD clinical trials, which generally use near-term change in FVC to determine therapeutic efficacy. In a recently published longitudinal FVC analysis, we found that change in FVC after satisfying many of the PPF features assessed here was highly heterogeneous. We found that CT progression of fibrosis, alone or

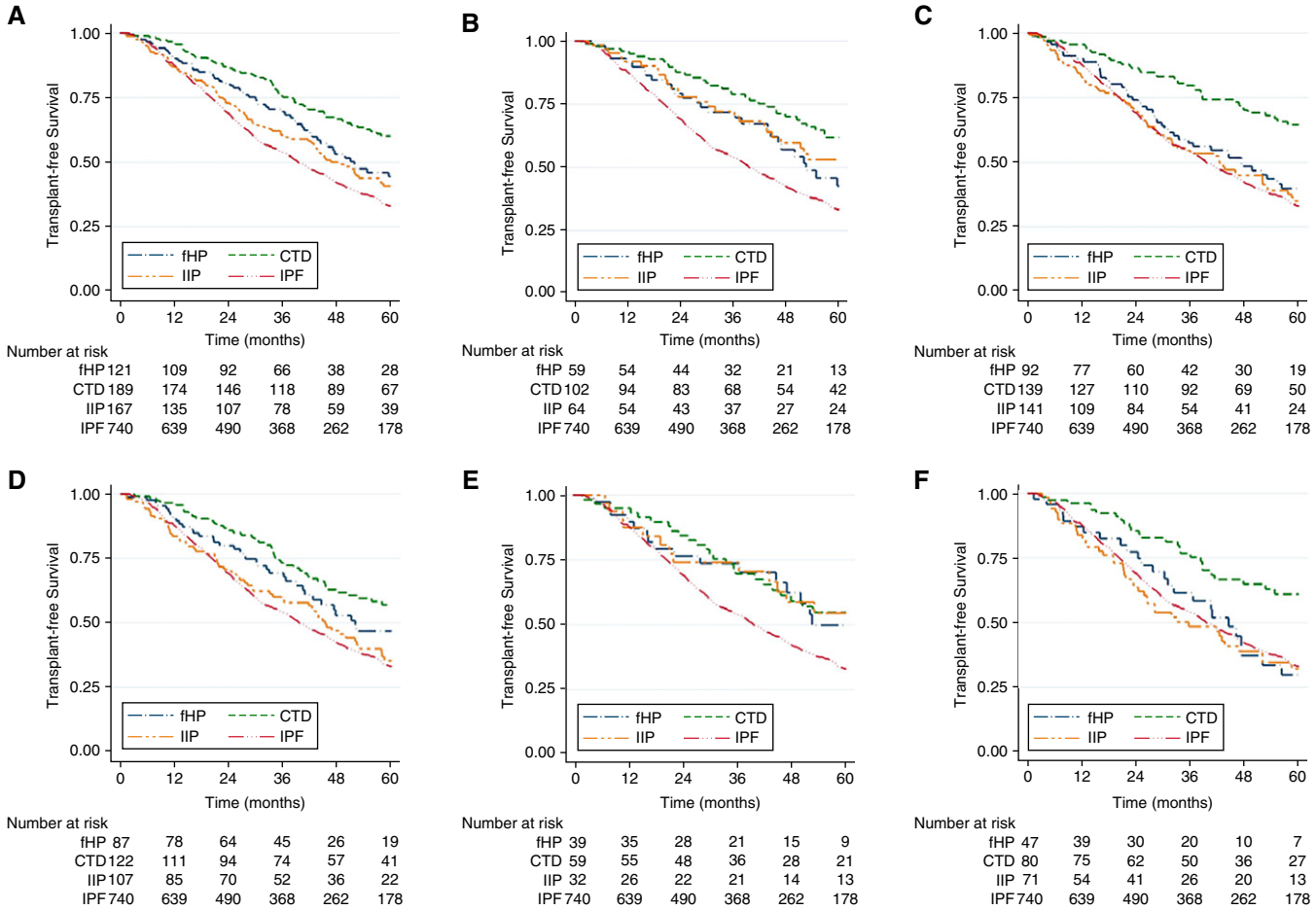


**Figure 4.** (*A–F*) Transplant-free survival among interstitial lung disease subgroups after satisfying validated progressive pulmonary fibrosis criteria in the absence of concurrent ⩾10% relative FVC decline, including (*A*) 5–9% relative FVC decline, (*B*) ⩾15% relative $D_{LCO}$ decline, (*C*) CT progression of fibrosis, (*D*) 5–9% relative FVC decline and worsening respiratory symptoms, (*E*) 5–9% relative FVC decline and ⩾15% relative $D_{LCO}$ decline, and (*F*) CT progression of fibrosis and worsening respiratory symptoms. fHP = fibrotic hypersensitivity pneumonitis; CT = computed tomography; CTD = connective tissue disease; IIP = idiopathic interstitial pneumonia; IPF = idiopathic pulmonary fibrosis.

in combination with physiologic or symptomatic worsening, strongly predicted subsequent FVC decline, while ⩾10% relative decline in percentage predicted FVC poorly predicted additional FVC decline. These findings corroborate prior findings in IPF (32, 33) but stand in contrast to a recent *post hoc* analysis of a PPF trial (20), leaving it unclear which criteria will optimally enrich clinical trial cohorts with patients most likely to progress during the trial follow-up period. This uncertainty does not appear to extend to survival, as categorical decline in FVC has long been known to reliably predict reduced survival across ILD subtypes (15–18), which we confirmed here.

Our study had several limitations. Although we sought to include only patients with non-IPF ILD, we cannot exclude the possibility that some patients without surgical lung biopsy had IPF. All patients were diagnosed by gold-standard multidisciplinary discussion, but we recognize that diagnostic agreement is variable among centers (34). Next, given the retrospective nature of this study, follow-up time varied by patient and cohort, precluding the assessment of each PPF feature over a standardized time frame. In addition, we relied on the medical record for symptomatic worsening and increasing fibrotic extent on CT. This may have introduced some degree of bias but was similar to methodology used in a recent PPF clinical trial (5). We elected not to assess symptomatic worsening as a stand-alone feature in this study given the high likelihood of ascertainment bias in the absence of objective lung function decline or high-resolution CT progression of fibrosis. This stand-alone, potential feature of PPF warrants prospective investigation using a validated symptom questionnaire. Finally, exclusion of those with <10% fibrotic involvement on high-resolution CT makes it unclear whether our results can be extrapolated to those with early ILD. Dedicated investigation of proposed PPF criteria in those with early ILD is warranted.

## Conclusions

Emergence of the PPF construct represents an important advancement for the field, as progression is highly variable among non-IPF ILDs, and identifying those with a PPF phenotype is of critical importance. In confirming that ⩾10% relative FVC decline

strongly predicted reduced TFS, and that six additional PPF criteria satisfied in the absence of ⩾10% FVC decline were also associated with reduced TFS, we have identified a set of PPF criteria that effectively identify a progressive phenotype in those with non-IPF ILD. Although these data should inform future iterations of consensus PPF criteria, they also strongly support prospective investigation of patients with non-IPF ILD to validate and expand on these findings. ■

**Author disclosures** are available with the text of this article at www.atsjournals.org.

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